Date:

1. a. Sebuah objek bergerak dengan percepatan $a(t) = -4$ cm^2/det dan pada $t=0$, kec $v=4$, dan $s=0$. Tentukan persamaan jarak s benda tersebut, dan lengkapi grafik $v(t)$, $a(t)$ dan $s(t)$.

$a(t) = \frac{dv}{dt} \rightarrow \int dv = \int a(t)\,dt$

$v(t) = \int -4 \; dt$

$v(t) = -4t + C$ → solusi umum

$v(t=0) = 4 \rightarrow 4 = -4(0) + C$

$C = 4$

$v(t) = -4t + 4$ → solusi khusus

$v(t) = \frac{ds}{dt} \rightarrow \int ds = \int v(t)\,dt$

$s(t) = \int (-4t+4)\,dt$

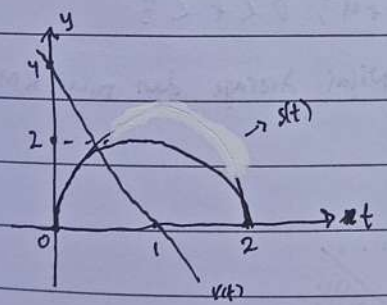
$s(t) = -2t^2 + 4t + C$ → solusi umum

$s(t=0) = 0 \rightarrow 0 = -2(0)^2 + 4(0) + C$

$C = 0$

$s(t) = 4t - 2t^2$ → solusi khusus

grafik $v(t)$, $a(t)$ dan $s(t)$



$v(t) = -4t + 4$

mencari $t \rightarrow y=0 \rightarrow 0 = -4t + 4$

$t = 1$

mencari $y \rightarrow t=0 \rightarrow y = -4(0) + 4$

$y = 4$

$s(t) = 4t - 2t^2$

mencari $t \rightarrow y=0 \rightarrow 0 = 4t - 2t^2$

$= -2t(t-2)$

$= -2t=0 \;|\; t-2=0$

$t=0 \quad t=2$

mencari $y \rightarrow t=0 \rightarrow y = 4(0) - 2(0)^2$

b. Arus rangkaian listrik (X) untuk memenuhi persamaan $i(t) = 4\sin(3t + \pi/6)$ Ampere, dan pada $t=0$, $q=0$ C. Tentukan persamaan muatan $q(t)$, dan hitung i pada $t=0$.

$i(t) = \frac{dq}{dt} \rightarrow \int dq = \int i(t)\,dt$

$q(t) = \int i(t)\,dt$

$q(t) = \int 4\sin(3t + \pi/6)\,dt$

$q(t) = -\frac{4}{3}\cos(3t + \pi/6) + C$ → soluli umum

$q(t=0) = 0 \rightarrow 0 = -\frac{4}{3}\cos(3(0) + \pi/6) + C$

$= -\frac{4}{3}\cos \pi/6 + C$

$= -\frac{4}{3}\cos 30^\circ + C$

$= -\frac{4}{3}\left(\frac{1}{2}\sqrt{3}\right) + C$

$= -\frac{4}{6}\sqrt{3} + C$

$C = \frac{4}{6}\sqrt{3} = \frac{2}{3}\sqrt{3}$

$q(t) = -\frac{4}{3}\cos(3t + \pi/6) + \frac{2}{3}\sqrt{3}$ → solusi khusus

$i(t=0) = 4\sin(3(0) + \pi/6)$

$= 4\sin \pi/6$

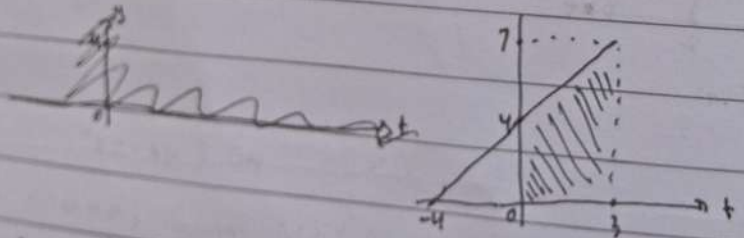
$= 4 \cdot \frac{1}{2} = 2$ Ampere

2. a. Suatu fungsi terdefinisi oleh $y = f(t) = t + 4;\ 0 < t < 3$

Sketsakan grafik $f(t)$, kemudian hitung Nilai Average dan nilai RMS y.

$f(t=0) = 0 + 4 = 4$

$f(t=3) = 3 + 4 = 7$



$f_{avg} = \frac{1}{b-a}\int_a^b f(t)\,dt$

$= \frac{1}{3-0}\int_0^3 t + 4\,dt$

$= \frac{1}{3}\left(\frac{1}{2}t^2 + 4t\right)\Big|_0^3$

$= \frac{1}{3}\left(\frac{1}{2}(3)^2 + 4(3)\right) - \left(\frac{1}{2}(0)^2 + 4(0)\right)$

$= \frac{1}{3} \cdot \left(\frac{9}{2} + 12\right)$

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$= \frac{1}{3}(16,5)$

$= 5,5$

$f_{rms} = \sqrt{\frac{1}{b-a}\int_a^b (f(t))^2\,dt}$

$= \sqrt{\frac{1}{3}\int_0^3 (t+4)^2\,dt}$

$= \sqrt{\frac{1}{3}\int_0^3 (t^2+8t+16)\,dt}$

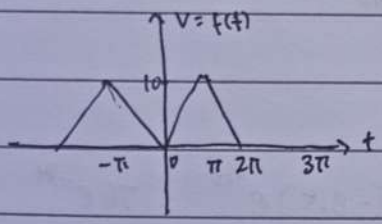
$= \sqrt{\frac{1}{3}\left(\frac{1}{3}t^3+4t^2+16t\right)\Big|_0^3}$

$= \sqrt{\frac{1}{3}\left(\frac{1}{3}(3)^3+4(3)^2+16(3)\right) - 0}$

$= \sqrt{\frac{1}{3}(9+36+48)}$

$= \sqrt{\frac{1}{3}(93)} = \sqrt{31} = 5,57$

b. Tegangan (V) ; $V = f(t)$ fungsi periodik dengan periode 2π seperti gambar berikut :



Definisikan $f(t)$ dalam satu periode, kemudian hitung nilai rms V

$f(t) = \frac{2A}{T}t$ untuk $0 \le t \le \frac{T}{2}$

$= \frac{2(10)}{2\pi}t = \frac{10}{\pi}t$ untuk $0 \le t \le \frac{T}{2}$

$f(t) = 2A - \frac{2A}{T}t$ untuk $\frac{T}{2} \le t \le T$

$= 2(10) - \frac{2(10)}{2\pi}t = 20 - \frac{10}{\pi}t$ untuk $\frac{T}{2} \le t \le T$

$f_{rms} = \sqrt{\frac{1}{T}\int_0^T f^2(t)\,dt}$

$= \sqrt{\frac{1}{2\pi}\int_0^{\pi}\left(\frac{10}{\pi}t\right)^2 dt + \int_{\pi}^{2\pi}\left(20-\frac{10}{\pi}t\right)^2 dt}$

$= \sqrt{\frac{1}{2\pi}\int_0^{\pi}\left(\frac{100}{\pi^2}t^2\right)dt + \int_{\pi}^{2\pi}\left(400-\frac{400}{\pi}t+\frac{100}{\pi^2}t^2\right)dt}$

$= \sqrt{\frac{1}{2\pi}\left[\frac{300}{\pi^2}t^3\right]_0^{\pi} + \left[400t-\frac{800}{\pi}t^2+\frac{300}{\pi^2}t^3\right]_{\pi}^{2\pi}}$

$= \sqrt{\frac{1}{2\pi}\left(\frac{300}{\pi^2}(\pi^3)-0\right) + \left(800\pi-\frac{1600}{\pi}\pi^2+\frac{600}{\pi^2}\pi^3 - \left(400\pi-\frac{800}{\pi}\pi^2+\frac{300}{\pi^2}\pi^3\right)\right)}$

$= \sqrt{\frac{1}{2\pi}(300\pi + (800\pi - 1600\pi + 600\pi - 400\pi + 800\pi - 300\pi)}$

$= \sqrt{\frac{1}{2\pi}(300\pi - 100\pi)}$

$= \sqrt{150-50}$

$= \sqrt{100} = 10$

3. Linear Dif Eq dengan input $p(t)$ dan output $q(t)$; $0{,}25\frac{dq}{dt} + 2q = v(t)$ dengan $q(0)=0{,}5$

Tentukan $q(t)$ dan $i(t)$ bila :

a. $v(t) = 20$

$0{,}25\frac{dq}{dt} + 2q = 20$

$\frac{dq}{dt} + 8q = 80$

$q(t) = e^{-\int 8\,dt} \int 80 e^{\int 8\,dt}\,dt$

$= e^{-8t} 80 \int e^{8t}\,dt$

$= 80 e^{-8t}\left(\frac{1}{8}e^{8t} + c\right)$

$= 10 + 80c e^{-8t}$ → solusi umum

$q(t=0) = 0{,}5 \rightarrow 0{,}5 = 10 + 80c(e^{-8(0)})$

$0{,}5 = 10 + 80c$

$c = \frac{0{,}5 - 10}{80} = -\frac{9{,}5}{80}$

$q(t) = 10 + 80\left(-\frac{9{,}5}{80}\right)e^{-8t}$

$= 10 - 9{,}5e^{-8t}$ → solusi khusus

$i(t) = \frac{dq}{dt} = \frac{d(10 - 9{,}5e^{-8t})}{dt} = (-8) \times (-9{,}5)e^{-8t} = 76e^{-8t}$

b. $v(t) = 20e^{-4t}$

$\frac{dq}{dt} + 8q = 20e^{-4t}$

$q(t) = e^{-\int 8\,dt} \int 20e^{-4t} \cdot e^{\int 8\,dt}\,dt$

$= 20e^{-8t} \int e^{-4t} \cdot e^{8t}\,dt$

$= 20e^{-8t} \int e^{4t}\,dt$

$= 20e^{-8t}\left(\frac{1}{4}e^{4t} + c\right)$

$= 5e^{-4t} + 20ce^{-8t}$ → solusi umum

$q(t=0) = 0{,}5 \rightarrow 0{,}5 = 5e^{-4(0)} + 20ce^{-8(0)}$

$0{,}5 = 5 + 20c$

$c = \frac{0{,}5 - 5}{20} = \frac{-4{,}5}{20}$

$q(t) = 5e^{-4t} + 20\left(\frac{-4{,}5}{20}\right)e^{-8t}$

$= 5e^{-4t} - 4{,}5e^{-8t}$ → solusi khusus

$$i(t) = \frac{da}{dt} = \frac{d}{dt}(5e^{-4t} - 4.5e^{-8t})$$

$$= 5(-4)e^{-4t} - 4.5(-8)e^{-8t}$$

$$i(t) = -20e^{-4t} + 36e^{-8t}$$